



STUDENT NAME: SHEHRYAR SHEHRYAR

STUDENT ID: P2952028

MODULE TITLE: AI FOR MOBILE ROBOTS

Shehryar Shehryar
DE MONTFORT UNIVERSITY, LEICESTER

Introduction:

Localisation is one of the core problems in mobile robotics, where a robot has to estimate its location and orientation within an environment known in advance. In this coursework, an already constructed 2D map is given to a TurtleBot3 robot and is required to localise itself while moving. The Robot Operating System (ROS) and the inbuilt Adaptive Monte Carlo Localisation (AMCL) is employed to calculate the robot's pose, i.e., its x position, y position and heading.

Localisation Technique:

The localization method used in this coursework is Adaptive Monte Carlo Localization (AMCL). AMCL is a probabilistic algorithm implemented through particle filters. The robot's potential poses on the map are given as particles set. Each particle is updated with odometry and its likelihood using laser scans.

The algorithm follows three main steps: Prediction, correction and resampling. At prediction, particles are propagated according to odometry readings. In the correction steps, laser scan data is compared with the map to assigned weighted to each particle. Finally, resampling ensures that particles with higher probability are retained more frequently. This approach allows the robot to correct odometry drift and maintain an accurate pose estimate.

Software Techniques:

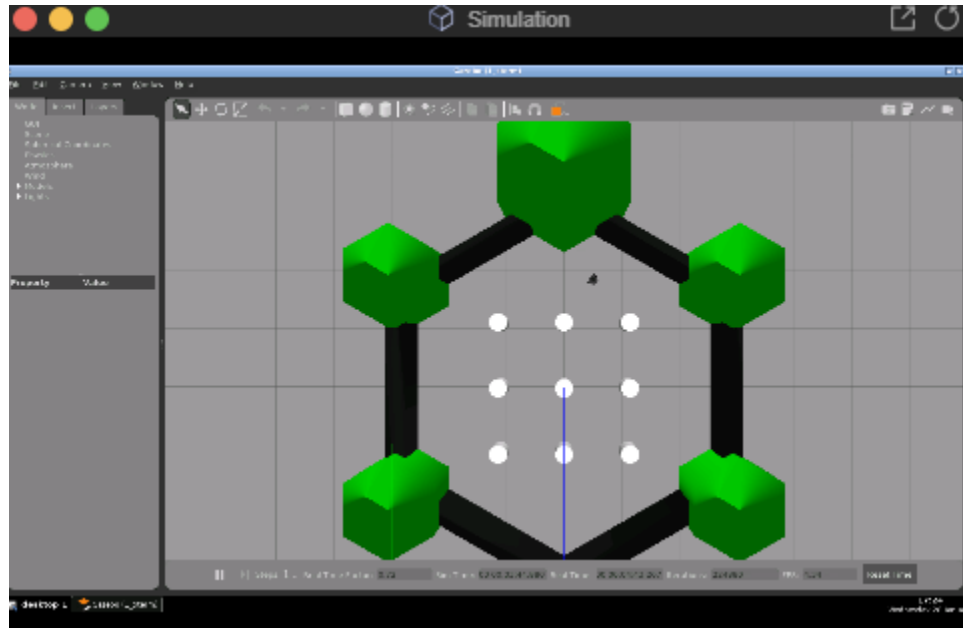
This coursework has been implemented in ROS Noetic and TurtleBot3 simulation environment. The robot and its environment are simulated in Gazebo. A previously created map that I create in assignment 1 (2D Map) is used and published on /map using the map_server node. AMCL subscribes to /scan and /odom topics being published localisation results.

RViz is used for visualization. The map, particle cloud, and estimated robot pose are displayed in real time. The Robot is moved either using keyboard teleoperation or an autonomous wandering robot node that publishes velocity commands to the /cmd_vel topic.

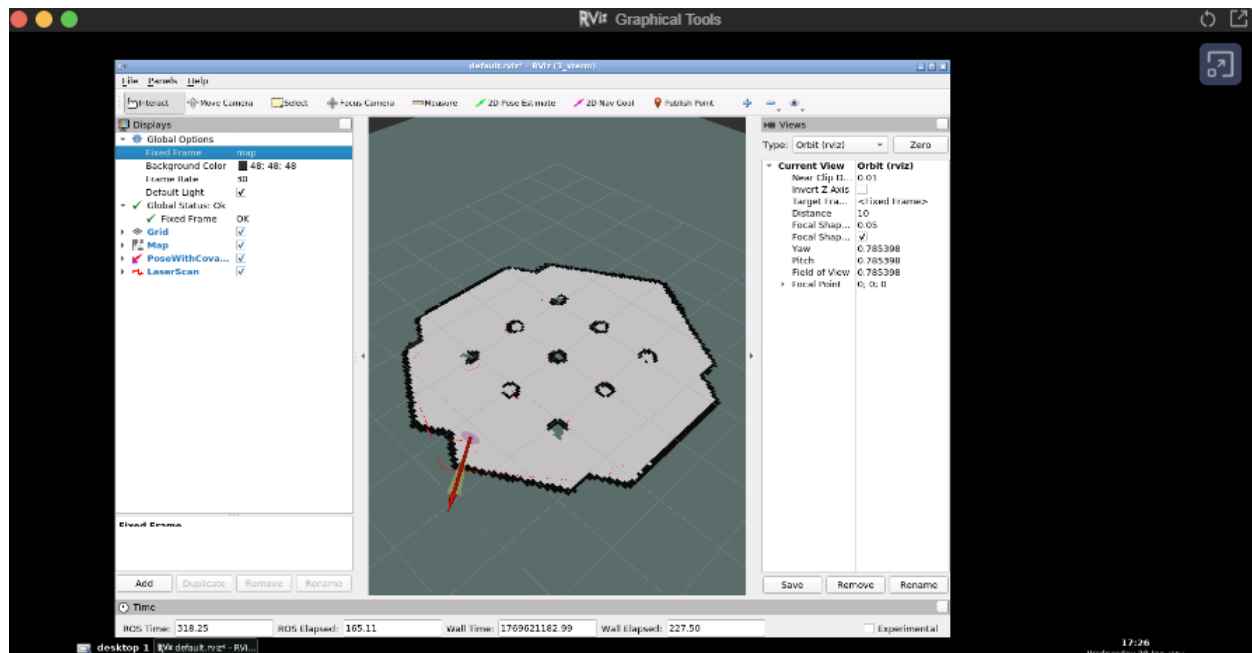
Testing and Results:

The localisation system was tested by moving the robot across simulated environments. A start pose was created by the use of the "2D Pose Estimate" tool in RViz. After this, the AMCL particles clustered around the actual robot's position. The particle cloud was evolving in time with the moving of the robot and was well matched to the map structure.

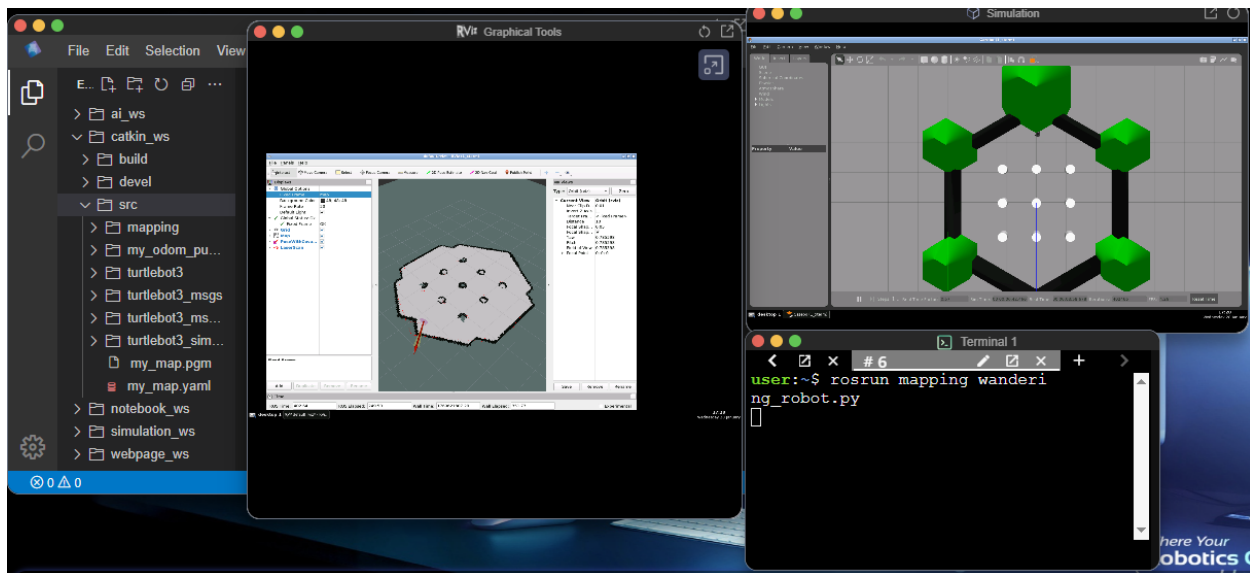
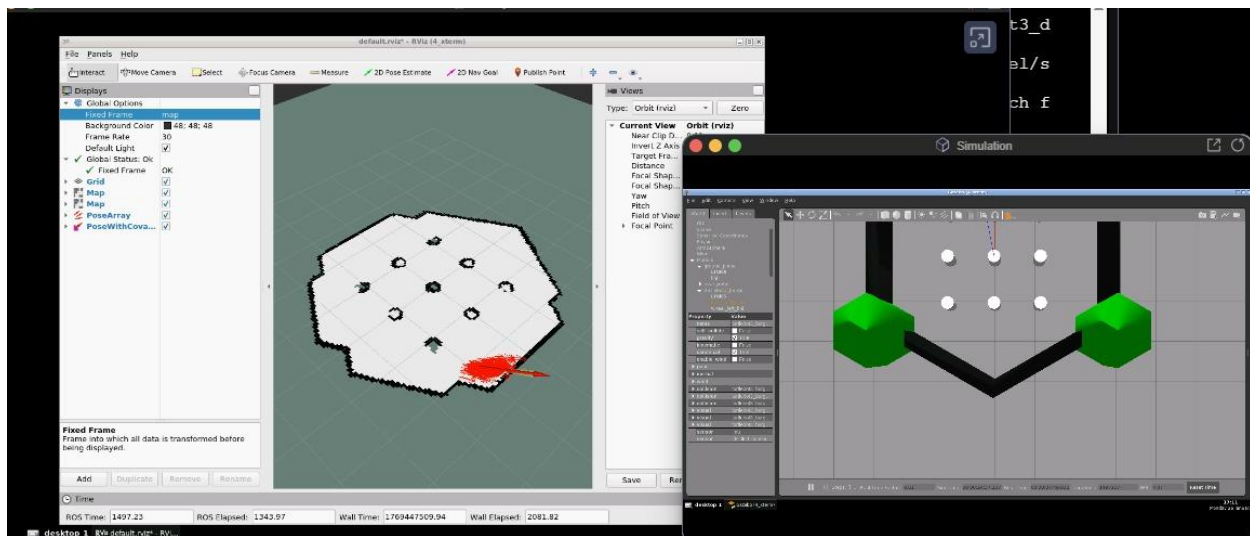
The estimated pose, which was published in the /amcl_pose topic updated several times per second, satisfying the assignment requirement. The localisation remained stable during both straight-line motion and rotations, showing that AMCL effectively combines odometry and laser data for accurate pose estimation.



FIGURE#1: Gazebo Map with moving Turtlebot3



FIGURE#2: Rviz with Laser Scan and Pose Arrow



FIGURE#3: Overall Project in Running Form

Conclusion:

This coursework is successfully demonstrated robot localisation using the AMCL package in ROS. The robot was able to accurately estimate its position and orientation within a known map while moving. The results show that AMCL provides a reliable and efficient localisation solution with minimal implementation effort.

References:

1. <https://uk.mathworks.com/help/ros/mapping-and-localization-applications.html>
2. <https://medium.com/@mansooralam129047/localization-in-robotics-for-mobile-robots-ec3ad31f99d4>
3. <https://link.springer.com/article/10.1007/s11370-023-00457-7>